

```
[path_1]
path = /directories/here
comment = something to comment
uid = nobody
gid = nobody
auth users = username
secrets file = /etc/rsyncd.scr
```

As you can see, this file has two sections—global parameters and path-specific settings. All file specifications like *motd* file (message of the day file), log file, *pid* file, lock file, etc, come under global specifications, while parameters that follow the *[path_1]* tag are path-specific settings (*[path_1]* is the name of that particular path). More than one such path can be set up in a configuration file. Though I have only specified relevant and important settings here, you can take a look at the *man* page if you want a complete list of path-specific and global configuration parameters. However, here is a brief explanation of some of them:

Parameter	Explanation
<i>path</i>	The physical directory in the filesystem to be made available.
<i>uid</i>	The user that should execute the actual transfer process.
<i>gid</i>	The group name or ID that accompanies the <i>uid</i> parameter.
<i>auth users</i>	The names of users, as specified in the <i>secrets</i> file, that are allowed to connect to this path. These do not refer to the actual users of the system.
<i>secrets file</i>	The name of the secrets file, specifying the user-names and passwords that are asked on connection to the client.

Now that *rsync* has been configured on the server, you can start the stand-alone daemon with the command *rsync --daemon*.

rsync client

The *rsync* server can now be used from the client with the following command:

```
rsync -avz host::path_1 /directory/at/destination
```

This command will copy all files from the remote module named *path_1* to the local destination folder on the client. This type of command is known as a *pull* command, while one with the remote path as destination is known as a *push* command. This will make it prompt for the password configured for the respective module, which can be bypassed by using the *--password-file* option (for use in scripts). Now, let's look at another example, of a local copy of files, with some of the optional switches:

```
rsync --verbose --progress --stats --compress --recursive
```

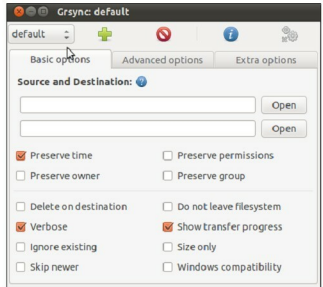


Figure 1: The Interface for Gsync

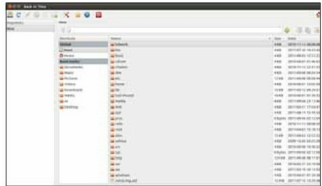


Figure 2: The GNOME view for Back In Time application

```
--times --perms --links --delete --exclude *.pdf /home/user/
Folder1/ /home/user/Folder2
```

Some of the options like *verbose*, *progress* and *stats* are self-explanatory. Compression (*--compress*) is one of the features *rsync* provides to speed up data transfers, which is actually useful when synchronising with remote hosts. The next few options specify the following:

- recursive — to recursively copy all sub-folders;
- times — to maintain previous timestamps
- perms — to keep the permissions intact
- links — to copy symbolic links as links (not to de-reference).

There are a number of other choices that you can use, like *--copy-links* and *--safe-links*; the first copies the linked directories/files instead of the links themselves, while the second altogether avoids links pointing outside the tree. The *--delete* option specifies that one wants to delete (earlier copied) files in the destination folder, if they have been deleted in the source directory since the previous synchronisation. Finally, *--exclude *.pdf* prevents all PDF files from being copied.